

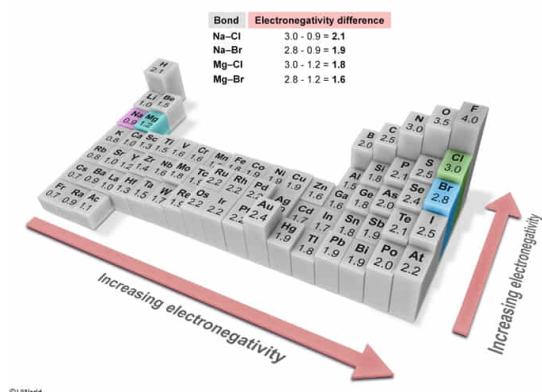
The bonds of four salts (MgBr_2 , NaCl , MgCl_2 , and NaBr) are evaluated for ionic character based on electronegativity differences between atoms. What is the expected order of these four salts if listed according to bond character from the least ionic character to the greatest ionic character?

- ☐ A. $\text{NaBr} < \text{NaCl} < \text{MgBr}_2 < \text{MgCl}_2$ [22%]
- ✓ ☒ B. $\text{MgBr}_2 < \text{MgCl}_2 < \text{NaBr} < \text{NaCl}$ [58%]
- ☐ C. $\text{NaCl} < \text{NaBr} < \text{MgCl}_2 < \text{MgBr}_2$ [10%]
- ☐ D. $\text{MgBr}_2 < \text{MgCl}_2 < \text{NaCl} < \text{NaBr}$ [7%]

Correct

59% answered correctly

Explanation:



An **ionic bond** between two atoms requires the transfer of **valence electrons** from one atom to the other, creating a cation and an anion. The **ionic character** increases as the **difference in electronegativity** between two atoms **increases**. Electronegativity tends to increase when moving across a period (row) from left to right and when ascending a group (column) on the periodic table, with some exceptions in the **transition metals**.

Comparing NaBr and MgBr_2 , NaBr has a higher electronegativity *difference* (and higher ionic character) because Na is further to the left of Br on the periodic table than Mg. Similarly, NaCl has a higher ionic character than MgCl_2 . Of the two higher-ranking sodium salts, NaCl has the greatest electronegativity difference (ionic character) as Cl is above Br in the halogens column.

Therefore, the expected order of these four salts if listed from the least ionic character to the greatest ionic character is $\text{MgBr}_2 < \text{MgCl}_2 < \text{NaBr} < \text{NaCl}$.

(Choice A) Mg is further to the right on the periodic table (closer to the halogens) than Na, giving a Mg-halogen bond a smaller electronegativity difference (less ionic character) than a comparable Na-halogen bond.

(Choice C) This is the order of the salts listed from the greatest to least ionic character (the opposite of that requested by the question). As such, the signs within this sequence are

incorrect. NaCl has *greater* ionic character than NaBr which is greater than MgCl_2 , and MgCl_2 is greater than MgBr_2 ($\text{NaCl} > \text{NaBr} > \text{MgCl}_2 > \text{MgBr}_2$).

(Choice D) Both NaCl and NaBr have a Na-halogen bond but Cl is more electronegative than Br, causing a greater electronegativity difference between Na and Cl.

Educational objective:

Electronegativity trends on the periodic table are useful to assess relative differences in electronegativity between elements. The difference in electronegativity between two bonded atoms is directly proportional to (and indicative of) the degree of ionic character of the bond between the atoms.

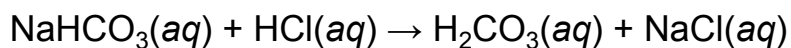
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Subject:
General Chemistry

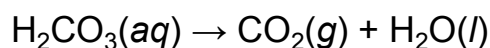
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System:
Interactions of Chemical Substances

Antacids act by functioning chemically as weak Brønsted-Lowry bases that neutralize excess stomach acid. Equations 1 and 2 describe the chemical behavior of the antacid sodium bicarbonate with stomach acid (HCl).



Equation 1



Equation 2

In Equation 1, carbonic acid (H_2CO_3) and sodium chloride form during a:

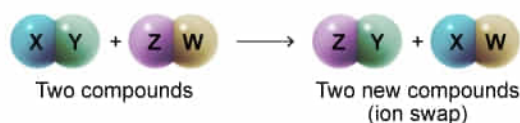
- ☐ A. single replacement reaction. [7%]
- ✓ ☒ B. double replacement reaction. [89%]
- ☐ C. combination reaction. [1%]
- ☐ D. decomposition reaction. [1%]

Correct

89% answered correctly

Explanation:

Double replacement reaction



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Chemical reactions can be categorized broadly into **five main types**: combination, decomposition, single replacement, double replacement, and combustion. **Double replacement** reactions involve an **exchange** of constituent **bonding partners** between two reacting compounds and result in the **formation** of **two new** product **compounds**. For **double replacement reactions** between two ionic species, the cations (or anions) of the two compounds simply exchange counter ions, and these exchanges frequently result in the formation of a new insoluble product (seen as a precipitate).

Equation 1 is a double replacement reaction because the bicarbonate anion (HCO_3^-) exchanges the sodium counter ion for an acidic proton from hydrochloric acid (forming carbonic acid, H_2CO_3), and the hydrochloric acid exchanges the acidic

proton for the sodium ion (forming sodium chloride, NaCl).

(Choice A) **Single replacement** reactions involve the reaction of a compound with a free *element* (a substance in its elemental form that is not bonded with other types of atoms) and the formation of a new compound and a new *element* as the products. None of the chemical species in Equation 1 are free elements.

(Choice C) **Combination** reactions follow the pattern of two or more chemical species reacting to combine and form *one* product species. Equation 1 has two products, and therefore does not follow this pattern.

(Choice D) **Decomposition** reactions follow the pattern of *one* reactant species breaking down into two or more product species (the opposite pattern of combination reactions). Equation 2 is a decomposition reaction, but Equation 1 does not follow this pattern.

Educational objective:

Chemical reactions can be categorized broadly into five main types: combination, decomposition, single replacement, double replacement, and combustion. Double replacement reactions involve the exchange of the bonding partners of two reactant compounds, forming two new product compounds.

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Subject:
General Chemistry

QID:401421
System:
Interactions of Chemical Substances